

DEPT OF ELECTRICAL & ELECTRONICS ENGINEERING

Program: T. Y. B. Tech ECE AI-ML Semester -6

Subject: Image Processing & Pattern Recognition (PC) (ECE3022B)

Experiment No.5

AIM: To perform region based segmentation of the given image

Introduction

Region-Based Segmentation: This technique groups pixels based on their inherent properties, such as color, intensity, or texture. By identifying areas with similar characteristics, region-based segmentation aims to create coherent and meaningful segments within an image.

Region growing technique:

In this segmentation, we grow regions by recursively including the neighboring pixels that are similar and connected to the seed pixel. We use similarity measures for regions with homogeneous gray levels. We use connectivity to prevent connecting different parts of the image. Similarity measures can be of different types: For the gray-scale image the similarity measure can be the different textures and other spatial properties, intensity difference within a region or the distance b/w mean value of the region.

Region Splitting

- Region growing starts from a set of seed points.
- An alternative is to start with the whole image as a single region and subdivide the regions that do not satisfy a condition of homogeneity.

Region Merging

- Region merging is the opposite of region splitting.
- Start with small regions (e.g. 2x2 or 4x4 regions) and merge the regions that have similar characteristics (such as gray level, variance).
- Typically, splitting and merging approaches are used iteratively.

Region merging techniques:

In the region merging technique, we try to combine the regions that contain the single object and separate it from the background.. There are many regions merging techniques such as Watershed algorithm, Split and merge algorithm, etc.

There are various region based segmentation methods namely Threshold based segmentation, Region growing, Region split and merge, Watershed segmentation, K-means clustering. Implement any two of these techniques.

Questions

Write down the algorithm of the implemented Region based segmentation technique.

Note

- **Write down the Expt Number, Title, PRN number, Name, and Date of performance in the first cell of the python code using markdown.**
- **Write down the “Result and Conclusion” in the cell next to the python code using markdown.**
- **Write down questions and answers in the last cell of the python code notebook using markdown.**